LAB 2 – AGILE BACKLOG CREATION & SPRINT SIMULATION IN JIRA

Customizable Subscription Box Scheduler

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Lab	Lab 2 – Agile Backlog Creation & Sprint Simulation in Jira
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1. Objective

The objective of this lab is to use Jira to convert functional requirements into an Agile backlog, organize the work using Epics and User Stories, estimate stories using story points, simulate sprints, and analyse sprint progress using the Burndown Chart.

2. Project Overview

Project Name: Customizable Subscription Box Scheduler

The system allows customers to select products, customize their subscription boxes, manage delivery schedules, manage subscriptions, handle payments, and receive relevant notifications.

3. Agile Backlog – Epics and User Stories

The backlog was organized into six major functional Epics. User Stories were written using the standard Agile format: "As a [user role], I want [goal], so that [benefit]."

Epic 1 – User Account Management

User Story	Story Description
User Registration	As a new customer, I want to create an account so that I can manage my subscription.
User Login	As a customer, I want to log in to my account so that I can securely access my subscription details.
Manage Profile	As a customer, I want to update my profile information so that my personal details remain accurate.

Epic 2 – Subscription Box Customization

User Story	Story Description
Browse Products	As a customer, I want to browse available products so that I can choose items for my subscription box.
Select Products	As a customer, I want to select products for my subscription box so that I can customize its contents.
Modify Box Contents	As a customer, I want to add or remove products from my box so that I can change my selection according to my preferences.

Epic 3 – Subscription Scheduling

User Story	Story Description
Select Subscription Frequency	As a customer, I want to select a delivery frequency so that I can receive my subscription box according to my preferred schedule.
Select Delivery Date	As a customer, I want to choose a delivery date so that my subscription box arrives when I need it.
Change Delivery Schedule	As a customer, I want to change my delivery schedule so that I can accommodate changes in my availability.

Epic 4 – Subscription Management

User Story	Story Description
View Subscription	As a customer, I want to view my active subscription details so that I know what products and schedule I have selected.
Pause Subscription	As a customer, I want to pause my subscription so that I can temporarily stop deliveries without cancelling it.
Cancel Subscription	As a customer, I want to cancel my subscription so that I can stop receiving subscription boxes when I no longer need them.
Resume Subscription	As a customer, I want to resume a paused subscription so that I can continue receiving my subscription boxes.

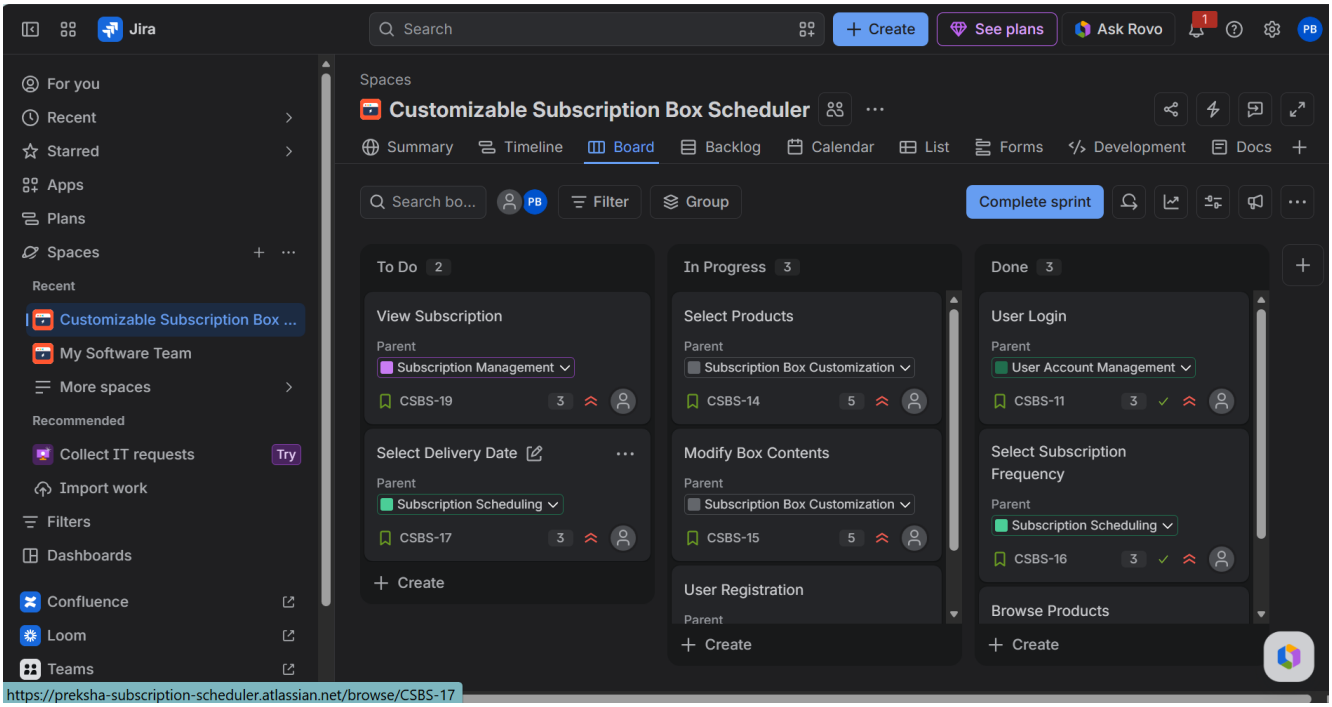
Epic 5 – Payment and Billing

User Story	Story Description
Add Payment Method	As a customer, I want to add a payment method so that I can pay for my subscription.
Process Subscription Payment	As a customer, I want my subscription payment to be processed securely so that my order can be confirmed.
View Billing History	As a customer, I want to view my payment history so that I can keep track of my subscription expenses.

Epic 6 – Notifications

User Story	Story Description
Delivery Notification	As a customer, I want to receive notifications about upcoming deliveries so that I know when to expect my subscription box.
Payment Notification	As a customer, I want to receive payment notifications so that I am aware of successful or failed transactions.
Subscription Update Notification	As a customer, I want to receive notifications when my subscription changes so that I remain informed about its status.

Screenshot 1 – Jira Backlog with Epics and User Stories



4. Story Point Estimation and Prioritization

Story points were assigned using Fibonacci-style values to represent relative complexity, effort, and uncertainty. Jira's available priority levels were used to distinguish the most important work.

User Story	Story Points	Priority
User Registration	3	Highest
User Login	3	Highest
Manage Profile	2	High
Browse Products	3	Highest
Select Products	5	Highest
Modify Box Contents	5	Highest
Select Subscription Frequency	3	Highest
Select Delivery Date	3	Highest
Change Delivery Schedule	5	High
View Subscription	3	Highest
Pause Subscription	3	High
Cancel Subscription	2	High
Resume Subscription	2	High
Add Payment Method	5	Highest
Process Subscription Payment	8	Highest
View Billing History	3	High
Delivery Notification	3	High
Payment Notification	3	High
Subscription Update Notification	3	High

Total backlog estimate: 69 story points.

5. Sprint 1 Simulation

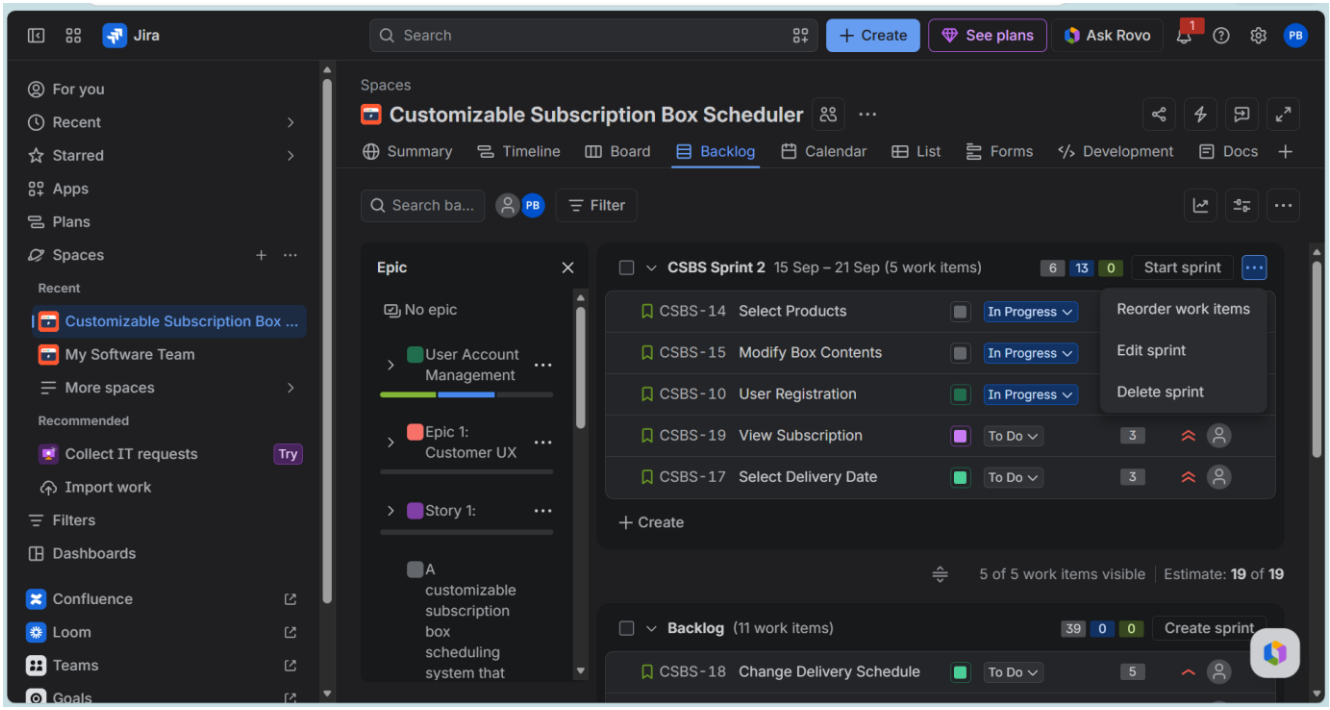
Sprint 1 was planned as a one-week sprint focused on account access, product browsing/customization, and basic subscription scheduling.

Sprint 1 Story	Points
User Registration	3
User Login	3
Browse Products	3
Select Products	5
Modify Box Contents	5
Select Subscription Frequency	3
Select Delivery Date	3
View Subscription	3

Sprint 1 planned capacity: 28 story points.

During the simulation, stories were moved through To Do, In Progress, and Done to represent the development workflow. Some work remained incomplete to make the sprint simulation realistic.

Screenshot 3 – Sprint 1 Active Sprint Board



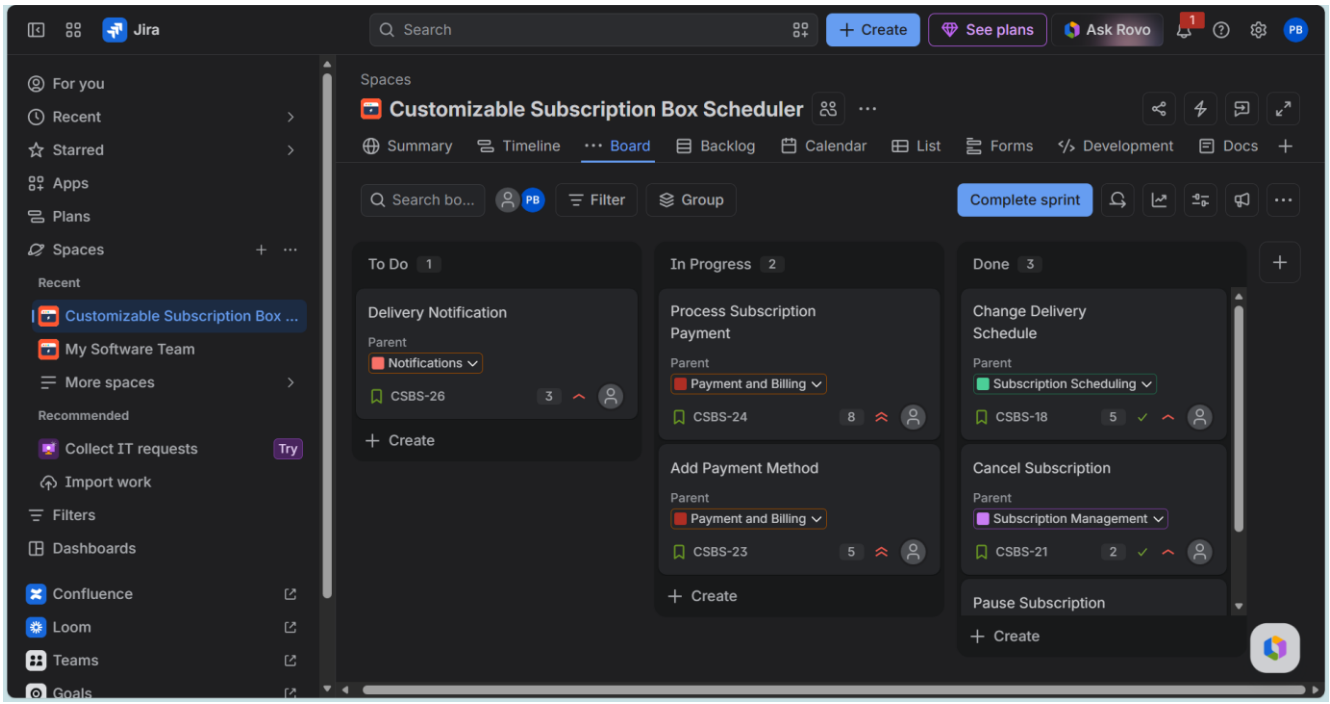
6. Sprint 2 Simulation

Sprint 2 continued the subscription management and payment/notification functionality. The Jira sprint was simulated by moving work items through the workflow and completing selected items.

Sprint 2 Story	Points
Change Delivery Schedule	5
Pause Subscription	3
Cancel Subscription	2
Add Payment Method	5
Process Subscription Payment	8
Delivery Notification	3

Note: The final Jira sprint/report reflects the actual simulated sprint state in Jira. The Burndown Chart should be interpreted from the recorded Jira sprint data.

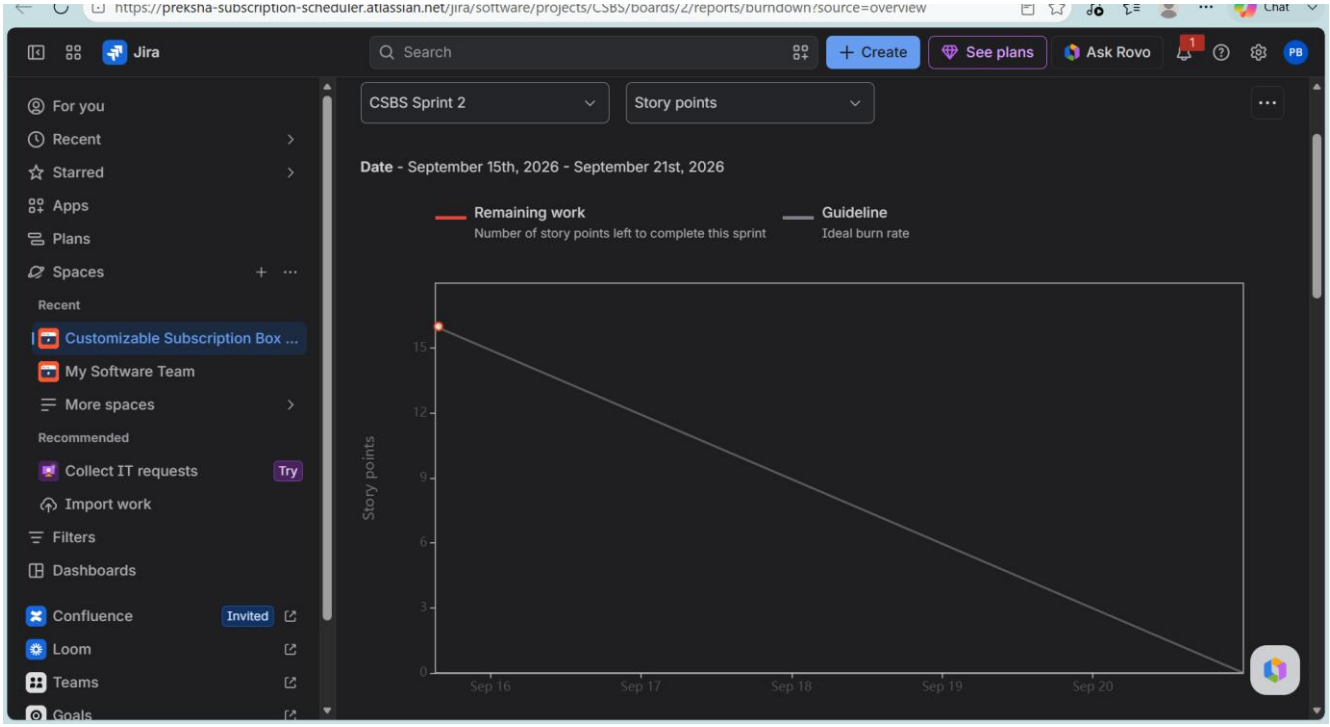
Screenshot 4 – Sprint 2 Active Sprint Board



7. Sprint Burndown Chart

The Sprint Burndown Chart shows the amount of remaining story-point work over the sprint period. It can be compared with the ideal guideline to understand progress and remaining work.

Screenshot 5 – Sprint Burndown Chart



8. Reflection and Analysis

1. Did your estimations reflect the actual effort?

The story-point estimates provided a reasonable relative indication of effort and complexity. Simple stories such as Cancel Subscription and Manage Profile were given lower estimates, while more complex functionality such as Process Subscription Payment was given a higher estimate. The sprint simulation also showed that estimation alone does not guarantee completion because actual progress is affected by workflow and team capacity.

2. Was the backlog well-prioritized?

Yes. Core customer-facing functionality such as registration, login, product browsing and selection, subscription scheduling, and payment processing was given the highest priority. Supporting functionality such as profile management, subscription changes, billing history, and notifications was assigned high priority. This ordering keeps the most important customer and transaction-related functionality ahead of secondary features.

3. How did your simulated sprint align with the original plan?

The sprint simulations generally followed the planned Agile workflow. Selected stories were placed into the sprint, worked through To Do and In Progress, and completed stories were moved to Done. Some stories remained incomplete, demonstrating that a sprint plan is a commitment based on expected capacity rather than a guarantee that every selected item will be completed.

4. What insights did the Burndown Chart give about team capacity?

The Burndown Chart showed how remaining story-point work changed during the sprint. A decrease in remaining work indicates completed items, while periods with little or no decrease indicate that work was still in progress or had not yet been completed. The chart therefore provides a visual indication of sprint pace, remaining workload, and whether the team's actual progress is close to the ideal burn rate.

9. Conclusion

This lab demonstrated the Agile backlog and sprint workflow using Jira. The project requirements were organized into Epics and User Stories, estimated with story points, prioritized, assigned to sprints, and tracked through the Jira workflow. The Sprint Burndown Chart was used to analyse progress and understand sprint capacity.